

SPECIFICATION OF ACOUSTIC PERFORMANCE OF ENTRY DOORS

I GENERAL

Provide minimum 50 mm thick solid core timber door with acoustical seals on all sides or a typical fire-rated metal door with air-tight gasket for Mechanical Equipment Rooms facing Office Services Corridor, Security Control Room & Management Offices, Network Operation Centre and Office.

This shall be read in conjunction with the relevant specifications and door schedule issued by the Architect. and the more stringent requirement shall be applied for the Contractor.

II ACOUSTICAL REQUIREMENTS

For solid core timber doors with acoustical seals, the doors shall be at least 50 mm thick solid wood (nominal density 650 kg/m³). Frames shall be solid wood, and the following are the acceptable acoustical seals with Zero International, Lorient, or equivalent, to achieve STC 35 or Noise Isolation Class (NIC) 28 in field.

For fire-rated metal door with gasket, air gap between door leaf and frame/floor should be no more than 5mm, and the following with Zero International, Lorient, or equivalent, are the acceptable acoustical seals to achieve STC 35 or Noise Isolation Class (NIC) 28 in field.

Position	Mfr's Type	Description
Head and Jamb	Zero #119 + #188 / #8144, or	Cushion Spring + Tear Drop / Batwing Self-Adhesive Perimeter Seal
	Zero #188 / #8144	Tear Drop / Batwing Self-Adhesive Perimeter Seal
Bottom Seal	Zero #369, or	Mortised Automatic Door Bottom
	Zero #361	Surface Mounted Automatic Door Bottom Drop Seal
Meeting Stiles	Zero #56 + 557	Adjustable Mortised Astragal

Noise Isolation Class (NIC) is a measure of the noise reduction between 2 spaces, as opposed to the Sound Transmission Class (STC) which is the transmission loss adjusted for room interior acoustical properties. The STC is more closely related to laboratory ratings, while NIC is a measure of the actual noise reduction experienced between existing spaces. The acoustic performance of door plus seal shall refer to drawings, door schedule, or acoustic report.

After installation of the door and to the satisfaction of the Project Manager, the Contractor shall perform acoustic test and confirm the design NIC rating is achieved. Field tests to be conducted in accordance with ASTM E336-90.

Any re-testing required due to product installation not meeting the acoustical performance will be performed by the Contractor at his own cost.

END OF SECTION